SOMI EdTech — Visual & Image Strategy (Final PRD)

Version: 2.0 — Merged Final **Date:** April 2026 **Product:** SOMI EdTech — AI Mentor Platform for CMA & NEET **Character:** Mama — Warm Telugu uncle mentor

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1. Core Philosophy

Every visual in SOMI must serve one of three purposes. If it doesn't, it doesn't ship.

Purpose	What It Does	Example
Memory	Helps student remember a concept longer	Mnemonic card, comparison diagram
Trust	Proves content is from official source	ICAI citation card, source reference
Motivation	Keeps student coming back daily	Streak fire, achievement badge, shareable card

No decorative visuals. No illustration for illustration's sake. Every pixel teaches, proves, or motivates.

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2. Mama — Dynamic Avatar System

2.1 Character Identity

- Warm, trustworthy Telugu uncle figure
- Semi-realistic animated style (Pixar-adjacent, not cartoon)
- Professional yet approachable — "SB" Institute badge on shirt
- Mama is the SOLE character in the CMA app (Kitty removed)

2.2 Expression Tag System

The backend returns an `expression_tag` with every Mama response. Flutter maps the tag to the correct avatar asset.

Expression Library (20 stickers, commissioned once):

Tag	Expression	Trigger
`EXPLAINING`	Hand gesture, open posture	Teaching a concept
`PROUD`	Wide smile, thumbs up	Student masters concept or passes quiz
`ENCOURAGING`	Gentle smile, leaning forward	Student fails but is trying
`SERIOUS`	Focused look	Exam tips, important warnings
`THINKING`	Hand on chin	Processing a question
`CONCERNED`	Slight frown, caring eyes	Student struggling or health alert
`DISAPPOINTED`	Gentle head shake	Streak broken, revision skipped
`SURPRISED`	Raised eyebrows	Student asks great question
`POINTING`	Finger pointing right	Directing to chapter or action
`LAUGHING`	Open laugh	Light moment, mnemonic joke
`HEALTH_REMINDER`	Holding supplement bottle	Medicine/vitamin reminder
`MOTIVATING`	Fist pump	Weekend goal push, target reminder
`GREETING_MORNING`	Waving, bright background	Daily morning session
`GREETING_NIGHT`	Calm, dim background	End of day summary
`FESTIVAL`	Festive clothing	Sankranti, Ugadi, Diwali greetings
`EXAM_DAY`	Confident posture	Special exam motivation
`CELEBRATING`	Both arms raised	Major milestone achieved
`COMPARING`	Two hands weighing	Comparison explanations
`WARNING`	Palm out, stop gesture	Common exam mistakes
`REVISING`	Flipping pages	Revision session

2.3 Context-Aware Backgrounds

Mama's background environment changes based on context:

Context	Background
Daytime study	Warm library with bookshelves
Late night (after 10 PM)	Dim desk lamp, cozy room
Exam approaching (< 7 days)	Exam hall aesthetic, focused lighting
Festival day	Festive decorations matching the occasion
Weekend	Slightly relaxed setting

2.4 Flutter Implementation

```
Backend Response:
{
  "text": "Ra Ravi, Going Concern ante...",
  "expression_tag": "EXPLAINING",
  "context": "daytime"
}

Flutter Mapping:
expression_tag → Asset path from local cache
context → Background overlay selection
```

- Store all 20 sticker PNGs/SVGs in app assets (not fetched per request)
- Small avatar (48x48) in chat bubbles
- Large sticker (200x200) for special moments (mastery, streaks, exam day)

Budget: ₹5,000–10,000 for full set from freelance illustrator.

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3. Automated Concept Visuals — JSON-to-Diagram Engine

3.1 Problem

4,625+ concepts cannot have manual illustrations. Cost and time make it impossible.

3.2 Solution

Claude generates diagram DATA as structured JSON at admin time. Flutter renders the diagram from JSON at display time. Zero manual design per concept.

3.3 Diagram Types

Type 1: Comparison Diagram Side-by-side cards for "this vs that" concepts.

```
json
```

```
{
  "type": "comparison",
  "title": "Going Concern vs Liquidation",
  "left": {
    "label": "Going Concern",
    "icon": "business_running",
    "points": [
      "Business continues operating",
      "Assets at cost price",
      "Like Appa's flour mill — still running"
    ]
  },
  "right": {
    "label": "Liquidation",
    "icon": "business_closing",
    "points": [
      "Business shutting down",
      "Assets at scrap/sale value",
      "Mill closing tomorrow — sell everything"
    ]
  }
}
```

Use cases: Cash vs Accrual, Revenue vs Capital Expenditure, Fixed vs Variable Cost, Asset vs Liability.

Type 2: Flow Chart Sequential steps for process-based concepts.

```json
{
  "type": "flowchart",
  "title": "The Accounting Cycle",
  "steps": [
    { "label": "Transaction", "icon": "receipt" },
    { "label": "Journal Entry", "icon": "book" },
    { "label": "Ledger Posting", "icon": "table" },
    { "label": "Trial Balance", "icon": "balance" },
    { "label": "Final Accounts", "icon": "report" }
  ],
  "direction": "horizontal"
}
```

Use cases: Accounting Cycle, Bill of Exchange lifecycle, Company Registration, Depreciation calculation.

Type 3: Relationship Map Pulled directly from the knowledge graph — shows how concepts connect.

```json
{
  "type": "relationship_map",
  "center": "Contract Act",
  "connections": [
    { "label": "Offer", "relation": "part_of", "strength": "strong" },
    { "label": "Acceptance", "relation": "part_of", "strength": "strong" },
    { "label": "Consideration", "relation": "part_of", "strength": "strong" },
    { "label": "Companies Act", "relation": "prerequisite_for", "strength": "me
  ]
}
```

Type 4: Formula Card Clean boxed formula with variable labels. Use styled text with superscripts/subscripts (NOT LaTeX — too heavy for Flutter).

```json
{
  "type": "formula",
  "title": "Straight Line Depreciation",
  "formula": "Depreciation = (Cost - Scrap Value) / Useful Life",
  "variables": {
    "Cost": "Original purchase price of asset",
    "Scrap Value": "Expected value at end of useful life",
    "Useful Life": "Number of years asset will be used"
  },
  "example": {
    "Cost": "₹1,00,000",
    "Scrap Value": "₹10,000",
    "Useful Life": "10 years",
    "Result": "₹9,000 per year"
  }
}
```

Type 5: Timeline Diagram For historical or sequential concepts.

```json
{
  "type": "timeline",
  "title": "Evolution of Accounting Standards in India",
  "events": [
    { "year": "1977", "label": "ASB established by ICAI" },
    { "year": "2006", "label": "Convergence with IFRS announced" },
    { "year": "2015", "label": "Ind AS implementation begins" },
    { "year": "2023", "label": "Ind AS mandatory for listed companies" }
  ]
}
```

3.4 Database Schema

```sql
concept_visuals table:
- concept_id (FK to concepts)
- has_diagram (boolean)
- diagram_type (comparison/flowchart/relationship/formula/timeline)
- diagram_json (JSONB — structured data for Flutter rendering)
- generated_at (timestamp)
```

3.5 Generation Pipeline

1. Admin triggers content generation for a chapter
2. Claude receives concept text + instruction to generate diagram JSON if visual would aid understanding
3. Claude returns `has_diagram: true/false` + `diagram_json` if applicable
4. Stored in `concept_visuals` table
5. Flutter reads JSON, renders using CustomPainter or pre-built widget templates
6. Not every concept needs a diagram — Claude decides

3.6 Flutter Rendering

- Build 5 reusable widget templates (one per diagram type)
- Each template takes the JSON and renders dynamically
- Consistent color palette from Design System (Section 9)
- All diagrams render as widgets, can be captured as images for sharing

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4. Mnemonic Cards

4.1 Types of Mnemonics

Acronym Mnemonics: Bold acronym + expansion + Tenglish memory hook.

- Accounting Concepts: G-A-A-P-C-M-M → "GAAP CMM" → "GAAP lo CMM company join ayyadu"
- Qualitative Characteristics: R-R-C-U → "RRCU" → "RR CU cheppadu"

Story Mnemonics (Tenglish): One-line story tying concept to a memorable Indian scenario.

- Going Concern → "Appa flour mill still running, so assets at cost price. Closing tomorrow ante scrap value lo record cheyyi"

Visual Icon Mnemonics: Simple icon paired with concept for instant recall.

- Prudence = 🛡️ Shield (always protect against losses)
- Consistency = 🔄 Repeat (same method every year)
- Materiality = 🔍 Magnifying glass (only report what matters)

4.2 Mnemonic Card Design

```
┌──────────────────────────┐
│  [Gradient background]   │
│                          │
│  G·A·A·P·C·M·M           │  ← Large bold acronym
│                          │
│  G — Going Concern       │
│  A — Accrual             │  ← Expansion
│  A — Accounting Period   │
│  P — Prudence            │
│  C — Consistency         │
│  M — Materiality         │
│  M — Money Measurement   │
│                          │
│  "GAAP lo CMM company    │
│   join ayyadu"           │  ← Tenglish hook in italics
│                          │
│  [Mama sticker: THINKING]│  ← Bottom right corner
└──────────────────────────┘
```

4.3 Generation Pipeline

1. Claude generates mnemonics per chapter at admin time
2. Prompt: "Create acronym mnemonic + one-line Tenglish story for these concepts: [list]"
3. Store in `concept_mnemonics` table
4. Serve as swipeable revision card deck in Flutter

4.4 Spaced Repetition Integration

- Mnemonic cards feed into spaced repetition scheduler
- SM-2 algorithm determines when to resurface each card
- Student mastered concept 10 days ago → schedule mnemonic review today
- Track: `last_reviewed`, `next_review_date`, `ease_factor` per card

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5. ICAI/ICMAI Source Citation Cards

5.1 Purpose

Every AI-simplified explanation must be anchored to the official ICAI/ICMAI textbook. This builds trust with students and parents — "SOMI teaches from the real textbook, not random internet content."

5.2 Design

- Teal/dark cyan background card
- "ICAI Definition:" prefix in bold white text
- Official definition text in clean white font
- "Source: Chapter X, Page Y, Para Z" in smaller text below
- Always positioned below Mama's explanation — never omitted

5.3 Rule

Every concept screen MUST show the ICAI source card. No exceptions. This is non-negotiable. It is the trust anchor that separates SOMI from every other edtech app teaching from random notes.

5.4 Data

Already available — 4,625 nodes parsed from ICMAI Paper 1 include `chapter`, `sub_chapter`, `book_page` fields.

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6. Health-Academic Correlation Engine

6.1 Vision

Mama doesn't just track study progress — he tracks the student as a whole person. Health affects academics. SOMI makes that connection visible.

6.2 The Visualization

- **Spider Chart:** Multi-axis chart showing Sleep, Energy, Nutrition, Exercise, Study Hours, Accuracy — all on one visual
- **Dual-Axis Line Graph:** Overlay health metrics (self-reported energy, supplement adherence) with academic metrics (accuracy, concepts mastered per day)

6.3 Data Collection (Phase 1 — Self-Reported)

Students don't need blood work. Start with simple daily check-ins:

Metric	Collection Method
Energy level	1-5 scale daily prompt from Mama
Sleep hours	Quick input: "Eppudu padukunnav?"
Supplement taken	Yes/No toggle reminder
Water intake	Optional tracker
Stress level	1-5 scale before exams
Exercise	Yes/No daily

6.4 Data Collection (Phase 2 — Medical Integration, Premium)

- DoctorC or lab report upload
- Extract Vitamin B12, D3, Iron, Cortisol levels
- Correlate with study accuracy over time
- Premium feature for later — NOT launch requirement

6.5 Health Alert Card

```
┌──────────────────────────┐
│ ⚠️ Health Alert          │
│                          │
│ [Mama: CONCERNED expression] │
│                          │
│ Ravi, last 4 days nee energy │
│ level 2/5 ki padipoyindi.    │
│ Same time lo accuracy 72% →  │
│ 58% ki drop ayyindi.         │
│                          │
│ Vitamin D supplement 3 days  │
│ miss ayyav. Take it today.   │
│                          │
│ [Mark as taken ✓]        │
└──────────────────────────┘
```

6.6 Correlation Logic

Pure application logic — no LLM needed:

```python
```

```
# If energy drops AND accuracy drops in same window
if avg_energy_last_5_days < 2.5 and accuracy_drop > 10%:
    trigger_health_alert(
        expression="CONCERNED",
        check_supplements=True,
        check_sleep=True
    )
```

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7. Progress Visualizations

7.1 Mastery Wheel (Syllabus Overview)

A 360-degree circular visualization of the entire syllabus:

- Each node = one concept
- Glowing bright node = mastered
- Dim node = attempted but not mastered
- Dark node = untouched
- Nodes arranged by chapter in concentric rings
- Student sees full syllabus mastery at one glance
- Tappable — tap any node to open that concept

7.2 In-Chat Progress Cards

When Mama reports progress, it renders as rich visual cards:

Overall Progress Card:

```
🎓 CMA Foundation - Paper 1

Overall: 45% ▰▰▰▰▰▰▰▰▱▱▱▱▱▱▱▱
Concepts: 2,081 / 4,625
Chapters: 5 / 12 complete

📅 At current pace:
Completion by May 28
Exam date: June 10
✅ On track
```

Chapter Progress Card:

- Chapter name + concept grid
- Each cell: green (mastered) / yellow (attempted) / red (failed) / grey (untouched)

- Tappable cells → open concept

Test Score Card:

```
┌──────────────────────────┐
│ 📝 Yesterday's Test      │
│                          │
│ Score: 12/20 (60%)       │
│ ✅ 12 Correct            │
│ ❌ 8 Wrong               │
│                          │
│ Wrong answers:           │
│ • Consideration [Review →]  │
│ • Going Concern [Review →]  │
│ • Accrual Basis [Review →]  │
│ ...5 more                │
│                          │
│ Trend: ↑ from 55% last test │
│ [Review all wrong answers →]│
└──────────────────────────┘
```

College Sync Card:

```
┌──────────────────────────┐
│ 🏫 College Sync Alert    │
│                          │
│ Your progress:  Ch 5 ▓▓▓▓▓▓▓▓▓▓│
│ College covered: Ch 8 ▓▓▓▓▓▓▓▓│
│ Gap: 3 chapters behind   │
│                          │
│ Mama's Plan: 2 extra concepts│
│ per day → catch up in 10 days│
│                          │
│ [Accept Mama's plan →]   │
└──────────────────────────┘
```

7.3 Animated Progress Bars

All progress bars use animated color gradients:

- 0-33%: Red gradient (needs work)
- 34-66%: Yellow/amber gradient (getting there)
- 67-100%: Green gradient (strong)

Animation: smooth fill from left on load, not instant.

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8. Gamification & Organic Growth Loop

8.1 Achievement Badges (Commission Once)

Badge	Name	Trigger
🔥	First Flame	Complete first concept
📚	Chapter Champion	Complete an entire chapter
🎯	Perfect Score	100% on a quiz
💪	Comeback Kid	Master a concept after 3+ failures
📅	7-Day Warrior	7 consecutive study days
📅	30-Day Legend	30 consecutive study days
🧠	Mnemonic Master	Reviewed all mnemonics in a chapter
⚡	Speed Demon	Completed quiz under time limit
🏆	Paper Complete	Finished an entire paper
🌟	Mama's Favorite	Special milestone set by Mama
🔄	Revision King	Completed full revision cycle
🏥	Health Hero	30-day supplement streak
📈	Rising Star	Improved accuracy 20%+ in a week
🎓	Exam Ready	Completed all papers for an exam level
🌍	Top 10%	In top 10% of all SOMI students

Budget: ₹3,000–5,000 for badge icon set from freelance designer.

8.2 Streak Visualization

- Fire icon that grows with streak length (small flame → large flame → blue flame at 30 days)
- Daily calendar grid: green = studied, grey = missed
- Streak record displayed prominently on home/chat screen

8.3 Shareable Achievement Cards

Purpose: Every achievement generates a shareable image. Students post on WhatsApp Status and Instagram Stories. Friends see SOMI → organic growth. Zero marketing cost.

Card Types:

Concept Mastery Card:

```
🎓 I mastered Going Concern
[Mama sticker: PROUD]
"Business continues → record
 at cost, not scrap value"
🔥 Day 12 Streak
SOMI EdTech | somi.app
[QR Code]
```

Test Score Card:

```
📝 Scored 95% on Accounts!
[Mama sticker: CELEBRATING]
SOMI EdTech | somi.app
[QR Code]
```

Streak Card:

```
🔥 30-Day Study Streak!
[Calendar grid visualization]
[Mama quote: "Consistency
 is the key to success"]
SOMI EdTech | somi.app
[QR Code]
```

Implementation:

- Flutter `RepaintBoundary` captures widget as image
- Native share intent (WhatsApp, Instagram, etc.)
- Always include: SOMI logo, app download link/QR code, student's first name
- QR code links directly to app download page

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9. Design System — The "Bentley Polish"

9.1 Color Palette

Role	Color	Hex	Usage
Primary	Warm Amber	`#F5A623`	Mama's header, buttons, highlights
Secondary	Teal	`#1A8A8A`	ICAI cards, trust elements
Success	Green	`#4CAF50`	Mastered, correct, streaks
Warning	Amber	`#FF9800`	Attempted, needs revision
Error	Red	`#E53935`	Failed, wrong answers
Background	Warm Cream	`#FFF8F0`	App background (eye-friendly for long study)
Text Primary	Dark Brown	`#3E2723`	Body text (warmer than pure black)
Text Secondary	Medium Brown	`#795548`	Subtitles, labels
Card Background	Soft White	`#FFFFFF`	Cards, bubbles
Chat Bubble (Mama)	Light Cream	`#FFF3E0`	Mama's messages
Chat Bubble (Student)	Light Amber	`#FFE0B2`	Student's messages

9.2 Typography

Element	Style
Headers	Bold, rounded sans-serif (e.g., Nunito Bold)
Body (Tenglish)	Regular weight, 16px, slightly larger for readability
Body (English translation)	Regular weight, italics in parentheses
ICAI Definitions	Clean serif or monospace for official feel
Key Terms	Bold + primary color highlight
Numbers/Scores	Large (24-32px), bold, accent color
Mnemonics	Extra bold (20px), gradient text

9.3 Card Design Rules

- All cards: 12-16px border radius
- Subtle shadow (elevation 2-4) for depth
- Consistent 16px internal padding
- No more than 2 action buttons per card
- Action buttons: primary color, full width at bottom

9.4 Chat Bubble Rules

- Mama's messages: cream background, left-aligned, avatar on left
- Student's messages: amber background, right-aligned
- System cards (progress, scores, achievements): full-width, no avatar, distinct background
- Action cards: tappable appearance with subtle border

9.5 Animation Rules

- Progress bars: smooth fill animation on load (300ms ease-out)
- Card entry: subtle slide-up + fade-in (200ms)
- Streak fire: gentle flicker animation (CSS/Lottie)
- Badge unlock: scale-up bounce effect
- Page transitions: smooth lateral slides
- No heavy animations on budget phones — respect ₹10-15K Android devices

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10. What NOT to Build

Don't Build	Why
Full illustration per concept	4,625 illustrations = too expensive, too slow. JSON diagrams solve this.
Video per concept	Defer until revenue justifies production.
3D animations or AR	Overkill for exam prep. Won't run on budget phones.
LaTeX rendering engine	Too heavy for Flutter. Styled text handles 95% of formulas.
Custom fonts beyond 1-2	Slow loading, increased app size. System fonts + one brand font max.
Heavy image backgrounds on every screen	Kills performance on budget Android phones.

Don't Build	Why
Kitty character	Removed. Mama is sole character in CMA app. Reserved for future.
Full DoctorC integration at launch	Start with self-reported health data. Medical integration is premium Phase 2.

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11. Budget Summary

Item	Cost	Frequency
Mama sticker set (20 expressions)	₹5,000–10,000	One-time
Achievement badge icons (15-20)	₹3,000–5,000	One-time
Chapter cover illustrations (12, optional)	₹5,000–8,000	One-time
Festival/seasonal cards (10)	₹2,000–3,000	Annual
Concept diagrams	₹0 (auto-generated)	—
Mnemonics	₹0 (auto-generated)	—
Progress visualizations	₹0 (code-rendered)	—
Shareable achievement cards	₹0 (code-rendered)	—
Health correlation charts	₹0 (code-rendered)	—
Mastery wheel	₹0 (code-rendered)	—
Total	**₹15,000–26,000**	**Mostly one-time**

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12. Implementation Priority

Week 1 — Foundation

- [ ] Commission Mama sticker set (20 expressions) — brief the illustrator
- [ ] Commission achievement badge icon set
- [ ] Implement Design System in Flutter (colors, typography, card templates)
- [ ] Build Mama chat bubble widget with expression tag mapping

Week 2 — Diagrams & Mnemonics

- [ ] Build 5 diagram widget templates (comparison, flowchart, relationship, formula, timeline)

- [ ] Add diagram JSON generation to admin content pipeline
- [ ] Build mnemonic card widget and generation pipeline
- [ ] Implement ICAI citation card component

Week 3 — Progress & Health

- [ ] Build mastery wheel visualization
- [ ] Build in-chat progress cards (overall, chapter, test score, college sync)
- [ ] Implement health check-in prompts from Mama
- [ ] Build health-academic correlation chart (spider chart)
- [ ] Build health alert card

Week 4 — Gamification & Growth

- [ ] Implement achievement badge system with unlock logic
- [ ] Build streak visualization (fire icon + calendar grid)
- [ ] Build shareable card generation (RepaintBoundary + share intent)
- [ ] Add QR code to shareable cards
- [ ] Build spaced repetition scheduler for mnemonic cards

Ongoing

- [ ] Festival cards per season
- [ ] New badge types as features grow
- [ ] Health integration Phase 2 (medical reports) — after revenue justifies

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13. Technical Architecture Summary

```
Admin Pipeline:
  Claude API → generates per concept:
    - V1/V2/V3 explanations
    - MCQ + trap option
    - Exam rubric
    - Mama's exam tip
    - diagram_json (if has_diagram = true)
    - mnemonic (acronym + Tenglish story)
  → All stored in Supabase

Student App (Flutter):
  Every Mama response includes:
    - text (Mama's words)
    - expression_tag (avatar selection)
    - context (background selection)
    - cards[] (rich UI cards to render)
    - actions[] (navigation/interaction triggers)

  Flutter renders:
```

```
    - Chat bubbles with dynamic Mama avatar
    - Diagram widgets from JSON
    - Mnemonic swipeable cards
    - Progress visualization cards
    - Health alert cards
    - Achievement badges
    - Shareable image cards

  No LLM call needed for:
    - Greetings, reminders, navigation
    - Progress reports, scores, streaks
    - Health reminders, supplement checks
    - Serving pre-generated content
    - Achievement unlocks
    - All diagram and mnemonic rendering
```

This document ensures that by launch, SOMI looks like it was built by a team of 50 — even though it's one founder and an AI board.